

A Technical Internship Report Submitted in Partial Fulfillment of the Requirements for the
DIPLOMA IN COMPUTER ENGINEERING
(Maharashtra State Board of Technical Education – MSBTE)

PROFORGE

AI-Driven Career Intelligence, Automated Resume Studio & Multi-Tenant Portfolio Suite

Submitted By

Dhruv R. Dubey

Enrollment No: 25112400240 | Class: TYCO - B

Under the Direct Supervision & Guidance of

Industry Mentor Prof. Harsh Tambade Founder & CEO, Elite Forums	College Mentor / Guide Ms. Mansi Patil Lecturer, Dept. of Computer Engg.
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VISHNU WAMAN THAKUR CHARITABLE TRUST'S

VIVA

INSTITUTE OF TECHNOLOGY

DEPARTMENT OF COMPUTER ENGINEERING

VIVA INSTITUTE OF TECHNOLOGY

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ABSTRACT

Industrial training is a compulsory and essential academic requirement mandated by the Maharashtra State Board of Technical Education (MSBTE) for the award of the Diploma in Computer Engineering. The primary objective of this internship is to bridge the gap between classroom theoretical concepts and live industrial software development practices, enabling diploma students to acquire practical hands-on competencies in modern programming languages, cloud architectures, database design, and real-time artificial intelligence workflows.

I completed my intensive 12-week industrial training program at Elite Forums, located in Vasai (East), Maharashtra, from 25th May 2026 to 15th August 2026. During this period, my daily work was organized into two distinct yet complementary operational segments: (1) On-Site Company Work & Training: Participating in company-assigned client software maintenance, internal testing, attending technical lectures on Python, JavaScript, and Supabase, and completing weekly coding evaluations; and (2) Independent Major Project Work: Designing, architecting, and fully implementing PROFORGE — An End-to-End AI-Powered Career Intelligence & Branding Suite.

ProForge was developed to resolve the widespread issue of career tooling fragmentation faced by students and job seekers. The platform unifies six full-featured AI-driven sub-suites: Remo AI (an intelligent resume builder featuring sub-second bullet refinement and a strict coordinate-based PDFKit rendering engine), Folio AI (an interactive portfolio generator producing zero-dependency downloadable ZIP packages), Talo AI (an ATS alignment checker comparing candidate profiles with job descriptions), Covo AI (a recruiter cold email and PDF cover letter studio), Liko AI (a LinkedIn personal branding engine), and Mali AI (an automated email scheduler with rich HTML templates).

The technical stack comprises React 18, Vite, Tailwind CSS, Node.js, Express REST APIs, Supabase PostgreSQL with JSONB schema structures, Groq SDK LPU inference (Qwen & Llama 3 models), Nodemailer with Zoho SMTP, and PDFKit. This report comprehensively documents the industrial hierarchy, equipment used, engineering methodologies, QA testing, digital security, 12-week task timelines, student learnings, and MSBTE curriculum outcomes achieved during the training.

ACKNOWLEDGEMENT

The successful completion of this 12-week industrial training and the development of the ProForge AI platform would not have been possible without the guidance, encouragement, and support of my mentors, teachers, and family. I take this opportunity to express my profound gratitude to everyone who contributed to this journey.

First and foremost, I express my sincere and deepest gratitude to my college mentor, Ms. Mansi Patil, Lecturer in the Department of Computer Engineering, VIVA Institute of Technology, Virar (East). Her continuous academic guidance, valuable suggestions, periodic monitoring, and patient feedback throughout the internship semester were instrumental in ensuring that this project strictly adheres to MSBTE guidelines and academic standards.

I am equally thankful to my industry mentor, Prof. Harsh Tambade, Founder and CEO of Elite Forums, Vasai (East), for providing me the opportunity to undergo this rigorous industrial training. His deep domain expertise in Generative AI, cloud infrastructure, and modern web application development, along with his mentorship on architectural patterns, helped me tackle complex full-stack engineering problems during the development of ProForge.

I also extend my heartfelt thanks to the Head of Department and all faculty members of the Department of Computer Engineering at VIVA Institute of Technology for their continuous encouragement and for providing the academic foundation required to excel in the software industry.

Finally, I thank the entire engineering and instructional team at Elite Forums for creating a supportive, collaborative learning environment, and my parents and friends for their unwavering moral support throughout this period.

Dhruv R. Dubey

Enrollment No: 25112400240

Class: TYCO - B (Third Year Computer Engineering)

VIVA Institute of Technology, Virar (East)

CHAPTER 1

ORGANIZATION STRUCTURE OF INDUSTRY & TEAM HIERARCHY

Elite Forums	
Founder & CEO Harsh Tambade	
General Manager Jeet Gharat	Chief Operating Officer (COO) Siddhant Mandlik
Project Manager Suchita Nigam	
Core Developers Team	Instructors & Mentors Team
Anshu Jaiswal Adarsh Pandey Mithilesh Vichare Yuvraj Singh	Shreya Mishra Prathamesh Jakkula Nandini Singh Shreya Mulik Shashank Singh

Elite Forums is an active IT services, software consulting, and technical upskilling company situated in Vasai (East), Palghar district, Maharashtra. The company follows a clean, modern organizational hierarchy designed for rapid software iteration and direct mentorship for diploma and degree engineering interns.

At the head of the organization is the Founder and CEO, Prof. Harsh Tambade, who drives strategic vision, cloud architecture, and corporate partnerships. The executive management consists of Jeet Gharat (General Manager) and Siddhant Mandlik (COO), who supervise daily operations, infrastructure, and industry outreach. Project workflows are managed by Suchita Nigam (Project Manager), who coordinates between the Developers Team (handling commercial client projects) and the Instructors Team (responsible for mentoring interns in Python, React, and Machine Learning).

This clear structure allowed me as an intern to receive direct feedback on my code, participate in sprint standups, and understand how commercial software engineering teams operate in practice.

CHAPTER 2

INTRODUCTION TO INDUSTRY (HISTORY, SERVICES & TRAINING MODEL)

ELITE FORUMS — IT SERVICES & CONSULTING

VASAI (EAST), PALGHAR, MAHARASHTRA – 401208 • ESTABLISHED 2023

Company Background & Mission

Elite Forums was established in 2023 with the goal of bridging the technical gap between academic engineering syllabi and the modern software industry. Operating with a lean team of 11 to 50 professionals, the firm provides tailored IT consulting, customized web solutions, and structured internship training programs for diploma and undergraduate engineering students across Maharashtra.

Core Business Offerings & Verticals

- Technical Training & Upskilling: Practical, project-based training programs in Full-Stack Web Development, Python Programming, Generative AI Integration, Cloud Databases, and Cybersecurity Fundamentals.
- Custom Software Development: Engineering bespoke web portals, database-backed management systems, and automated REST API pipelines for small and medium-sized businesses.
- IT Advisory & Cloud Migration: Helping traditional businesses migrate local workflows to modern cloud platforms like Supabase, AWS, and modern serverless platforms.

The Industrial Training Model for Diploma Students

As a diploma student in Computer Engineering following the MSBTE curriculum, the industrial training at Elite Forums provided a practical learning curve. Rather than merely listening to lectures, students are required to write production-grade code, participate in daily standups, use Git for version control, and present their work in weekly sprint reviews. This practical exposure helped me develop strong confidence in full-stack web technologies.

CHAPTER 2 (CONTINUED)

DUAL SCOPE: ON-SITE COMPANY WORK VS INDEPENDENT PROFORGE PROJECT

Clarification of Daily Work & Project Independence

To provide complete academic transparency for MSBTE evaluation, my 12-week internship work was categorized into two distinct operational components: On-Site Company Assigned Tasks and Independent Capstone Project Engineering (ProForge).

Category / Dimension	1. On-Site Company Work & Training	2. Independent Project (ProForge AI)
Primary Nature	Structured company training, internal tasks, and learning sprints.	Fully independent design, architecture, coding, and testing.
Daily Activities	Attending technical modules, completing weekly coding assignments, reviewing client UI fixes, and debugging sample databases.	Building the 6 ProForge sub-suites (Remo, Folio, Talo, Covo, Liko, Mali) from scratch using React, Express, and Groq SDK.
Supervision	Instructors Team & Project Manager (Suchita Nigam) at Elite Forums.	Mentored by Prof. Harsh Tambade (Industry) and Ms. Mansi Patil (College Guide).
Intellectual Ownership	Internal exercises and client tickets owned by Elite Forums.	Independent student project built by Dhruv R. Dubey for MSBTE diploma fulfillment.
Key Outcomes	Mastered Git, JavaScript ES6, React components, Node.js REST APIs, and Supabase.	Delivered an end-to-end, multi-tenant career intelligence suite with 100% strict A4 vector PDF engine.

This dual-track approach ensured that I received solid foundational training from the company while also challenging myself to independently build a production-grade software platform that solves genuine student recruitment problems.

CHAPTER 3

TYPES OF MAJOR EQUIPMENT, HARDWARE & SOFTWARE TOOLS USED

Building a modern web application requires a well-structured set of development tools, operating environments, and cloud infrastructure. During the internship, I worked with the following primary hardware and software stack:

Hardware Workstation Configuration

- Processor: Intel Core i5 / i7 Multi-Core Processor (x64 Architecture).
- System Memory (RAM): 16 GB DDR4 RAM (essential for running Vite dev servers, Node.js runtimes, and local browsers simultaneously).
- Storage: 512 GB NVMe Solid State Drive for high-speed file compilation and fast build times.
- Operating System: Microsoft Windows 11 64-bit with PowerShell and Git Bash terminals.

Software Development Stack & Libraries

- Code Editor & IDE: Visual Studio Code with ESLint, Prettier, and Tailwind CSS IntelliSense extensions.
- Frontend Core: React 18 (Component Architecture), Vite (Lightning-fast HMR build tool), Tailwind CSS v3 (Utility Styling), Lucide React (Vector Icons).
- Backend Framework: Node.js (V8 JavaScript Runtime), Express.js (RESTful API pipeline), CORS, Dotenv.
- Database & Storage: Supabase (Cloud-hosted PostgreSQL engine with JSONB column support and Row Level Security).
- AI Inference Engine: Groq SDK (High-speed LPU inference using Qwen 2.5 and Llama 3 models for sub-second text analysis).
- Document Compiler: PDFKit (Server-side coordinate-based vector PDF generator configured for strict A4 physical boundaries).
- Client-Side Archiving: JSZip (Client-side ZIP package generator for exporting self-contained HTML/CSS portfolio packages).
- Email Dispatch: Nodemailer with Zoho SMTP transporter for sending 6-digit OTP verification codes and scheduled outreach campaigns.

CHAPTER 3 (CONTINUED)

DETAILED SPECIFICATIONS & MAINTENANCE SCHEDULE

Tool / Technology	Specification / Version	Approx. Cost	Specific Purpose in Project	Routine Maintenance
Dev PC	Intel i5/i7, 16GB RAM, 512GB SSD	₹55,000	Local coding, server execution, multi-tab browser testing	OS updates, disk cleanup, cooling vent cleaning
React 18 & Vite	v18.2 / Vite v5.0 (SPA Architecture)	Open Source	Dynamic UI dashboards, theme switches, state management	npm dependency updates, bundle tree-shaking
Node.js & Express	Node v20 LTS / Express v4.19	Open Source	REST API routing, token validation, PDFKit streaming	Handling unhandled rejections, monitoring event loops
Groq SDK (AI)	Groq Cloud SDK / LPU Architecture	Free / Tiered	Sub-second AI resume parsing, ATS scoring, bullet rewrites	Configuring rate-limit retries and fallback models
Supabase (DB)	PostgreSQL 15 with JSONB support	Free Tier	User profile persistence, OTP tracking, login history	Automated daily snapshots, schema migrations
PDFKit Engine	PDFKit v0.15 (Server Vector PDF)	Open Source	Generating strict A4 print resumes & cover letters	Dynamic coordinate math checks, margin auditing
Zoho SMTP	TLS Port 587 / SSL Port 465	Free / Business	Dispatching 6-digit OTP codes and outreach emails	Monitoring bounce rates, verifying SPF/DKIM records

By carefully selecting and maintaining these tools, the platform achieved high operational stability, zero licensing costs for development, and rapid build times throughout the 12-week development lifecycle.

CHAPTER 3 (CONTINUED)

GROQ LPU AI ENGINE, MULTI-MODEL FALLBACK & ASYNC JAVASCRIPT

Asynchronous JavaScript & Non-Blocking Event Loop

In modern web applications, operations like querying a remote database, calling external AI APIs, or rendering multi-page PDF documents introduce network and compute latency. Because Node.js operates on a single-threaded event loop, running these tasks synchronously would freeze the entire server, blocking requests from all other users.

To prevent server blocking, ProForge utilizes asynchronous JavaScript (async/await and native ES6 Promises) across all backend controllers. When a user requests resume analysis, the server offloads the AI call to the event loop worker pool, freeing the main thread to handle other incoming requests without delay.

Multi-Model Fallback Cascade for 99.9% AI Availability

Third-party AI APIs can occasionally suffer from rate-limiting (HTTP 429 errors), network timeouts, or service downtime. To ensure uninterrupted user experience, I designed a multi-model fallback cascade inside backend/groqClient.js. If the primary model is busy, the request cascades automatically to backup models before falling back to a deterministic rule-based extractor:

Step 1: Primary Model	Step 2: Secondary Fallback	Step 3: Tertiary Fallback	Step 4: Deterministic Guard
qwen/qwen3.8-27b Ultra-fast 27B model for quick JSON structuring.	openai/gpt-oss-120b High-capacity model if Qwen is rate-limited.	openai/gpt-oss-20b Lightweight model if high-capacity is busy.	Rule-Based Extractor Offline regex parser so UI never shows blank error.

This resilient architecture guarantees that students and job seekers can always generate, refine, and download their career materials even during upstream cloud API disruptions.

CHAPTER 4

PROCESSES, AGILE METHODOLOGIES & DIGITAL ASSET HANDLING

Software Engineering Processes in IT Industry

In the software engineering and IT domain, the term 'manufacturing process' refers to the Software Development Life Cycle (SDLC)—the structured procedure through which software requirements are gathered, designed, coded, tested, and deployed. At Elite Forums, we adhered to the Agile-Scrum framework, dividing the 12-week internship into six iterative two-week sprint cycles.

Agile Scrum Lifecycle Steps Followed

- **Sprint Planning:** At the beginning of each two-week cycle, sprint backlogs were created with specific feature goals (e.g., Sprint 3: Building the Remo AI Resume Editor and connecting Supabase).
- **Daily Standups:** 15-minute daily synchronization meetings to discuss completed tasks, planned work for the day, and any technical blockers encountered.
- **Feature Branching & Code Reviews:** Writing modular code on dedicated Git branches and submitting Pull Requests (PRs) on GitHub for review by senior developers before merging.
- **Sprint Demonstrations & Retrospectives:** Demonstrating functioning prototypes at the end of each sprint to gather constructive feedback and refine user interface interactions.

Digital Material & Asset Handling Procedures

Unlike manufacturing plants handling raw materials, software engineering requires careful governance of digital assets:

- **Source Code Management:** Managed through GitHub repositories with branch protection rules to prevent accidental overwrites on the main branch.
- **API Payload Contracts:** Standardized JSON request and response schemas shared across frontend React components and backend Express routes.
- **Static Assets & Fonts:** Vector SVG icons (Lucide React) and Google Web Fonts packaged locally to ensure lightning-fast rendering without external CDN delays.

CHAPTER 4 (CONTINUED)

MODULAR COMPONENT HIERARCHY & CSS DESIGN TOKENS

Component-Based Architecture in React 18

A key software engineering principle emphasized during my internship was component modularity. Instead of writing massive, monolithic web pages, the frontend UI was broken down into small, reusable, single-responsibility components:

- `Navbar.jsx`: Global navigation header featuring navigation tabs, quick-action buttons, and the active session indicator.
- `ThemeToggle.jsx`: Floating control panel enabling users to switch between 10 color themes and dynamic typography fonts with a single click.
- `TextBox.jsx`: Clean input textarea with auto-resizing, character counting, and pre-filled prompt presets for quick user onboarding.
- `OTPInput.jsx`: 6-digit verification component with automatic focus progression, backspace navigation, and clipboard paste detection.
- `AmbientBackground.jsx`: GPU-accelerated background layer providing subtle visual ambiance tailored to each active theme.
- `PasswordStrength.jsx`: Dynamic security meter evaluating password entropy in real-time.

Design Tokens & Multi-Theme Synchronization

ProForge supports 10 distinct aesthetic themes (including Cyberpunk Neon, Clean Teal, Midnight Cosmic, Emerald Executive, and Classic Monochrome). Rather than hardcoding colors into individual components, styling is managed through CSS Custom Properties (Design Tokens) defined in `index.css`. When a user selects a new theme, the root DOM element updates its data attribute, instantly propagating color transitions across all 6 sub-suites with smooth 200ms ease transitions.

CHAPTER 4 (CONTINUED)

UNIDIRECTIONAL FULL-STACK DATA FLOW & SUPABASE JSONB SCHEMA

Unidirectional Data Flow Architecture

ProForge follows a predictable, unidirectional data flow architecture that guarantees data consistency across the entire application:

1. Candidate Input: The user enters raw career notes or selects sample data in the frontend React view.
2. API Dispatch: The frontend sends a structured POST request to the backend Express route (/api/analyze).
3. AI Inference: The backend passes the text to Groq SDK, returning a validated JSON profile containing structured arrays for experience, education, skills, and projects.
4. Cloud Persistence: The profile payload is saved to Supabase PostgreSQL in a flexible JSONB column.
5. Vector Compilation: When exporting, the backend PDFKit engine reads the JSON profile, applies exact A4 coordinate math, and streams the binary PDF to the user's browser.

Supabase Database Schema Design

Traditional relational tables with rigid columns make it difficult to accommodate diverse resume structures (some candidates have 5 projects and 2 degrees; others have 10 certifications and no formal degree). By utilizing PostgreSQL's native JSONB (Binary JSON) columns in Supabase, ProForge stores nested, variable-length career structures while retaining fast indexed lookups and transactional safety.

CHAPTER 5

QUALITY ASSURANCE, UNIT TESTING & WEEKLY ASSESSMENTS

Quality Assurance (QA) Philosophy at Elite Forums

In commercial software development, code is only as good as its test coverage and reliability. Elite Forums maintained a strict quality assurance culture where interns were trained to write defensive code, handle edge cases, and participate in weekly technical evaluations.

Evaluation Formats Conducted During Training

1. Weekly Technical Quizzes (MCQs): Timed evaluations covering JavaScript ES6 concepts (closures, Promises, event loops), React lifecycles, and database normalization.
2. Live Coding Challenges: Timed problem-solving sprints where interns implemented stateful algorithms, debounced search bars, and recursive JSON tree traversers under mentor supervision.
3. REST API Testing with Postman: Creating automated Postman collections to test all backend endpoints against positive and negative test cases (e.g., missing auth tokens, malformed JSON bodies, SQL injection attempts).
4. Technical Slide Presentations (PPTs): Weekly slide deck presentations where interns explained system architectures, API workflows, and database schemas to mentors and peers, building professional communication skills.

This rigorous testing and assessment framework ensured that ProForge was built with high reliability, zero critical crashes, and clean, readable code.

CHAPTER 5 (CONTINUED)

PDFKIT A4 COORDINATE MATH & ATS SCORING VALIDATION

The Challenge of Strict A4 Physical Page Boundaries

A major technical challenge encountered during testing was ensuring that generated resumes fit cleanly onto standard A4 paper dimensions (595.28 x 841.89 points / 210 x 297 mm). When users added long job descriptions, traditional browser print engines often cut lines in half or pushed single lines onto awkward second pages.

Defensive PDFKit Coordinate Calculation Strategy

To resolve this, I implemented an automated coordinate calculation algorithm inside `backend/services/pdfService.js`:

- **Height Measurement:** Using `doc.heightOfString()` to pre-calculate the vertical pixel height of every text block and bullet point before drawing.
- **Dynamic Page Break Triggers:** If rendering a new experience entry would exceed the safe bottom margin (780 pt), the engine automatically issues a `doc.addPage()` command and redraws the section header.
- **Synchronized Two-Column Layout:** Enforcing independent Y-coordinate tracking for the left main content column (340 pt width) and the right sidebar (180 pt width), preventing column collapse.

ATS Scoring Engine Validation (Talo AI)

We validated Talo AI's ATS scoring algorithm by testing 50+ diverse resume profiles against real job descriptions across Software Engineering, Data Analysis, and Web Development roles. The test suite verified that keyword extraction accurately identified matching vs missing competencies and produced actionable improvement suggestions.

CHAPTER 5 (CONTINUED)

CROSS-BROWSER COMPATIBILITY, DIRECTWRITE & OTP AUTH SECURITY

Cross-Browser Compatibility & Typography Rendering

To guarantee an identical user experience across all devices and platforms, the ProForge web application was rigorously tested across multiple modern web browsers:

- Google Chrome & Microsoft Edge (Chromium Engine): Tested hardware acceleration, GPU canvas rendering, and fast HMR performance.
- Mozilla Firefox (Gecko Engine): Verified CSS custom property inheritance, smooth scrolling behaviors, and font antialiasing.
- Apple Safari (WebKit Engine): Tested mobile responsive layouts, touch gesture responsiveness, and backdrop-filter glassmorphism effects.
- DirectWrite & Font Smoothing: Configured -webkit-font-smoothing: antialiased and text-rendering: optimizeLegibility to prevent blurred typography on Windows displays.

OTP Authentication & Session Integrity Testing

We conducted thorough security testing on the Zoho SMTP 6-digit OTP verification flow. The test suite verified that: (1) OTPs automatically expire after 10 minutes, (2) Brute-force attempts are blocked after 5 failed tries, (3) Used OTPs cannot be replayed, and (4) Unauthorized API requests without valid JWT session headers return immediate HTTP 401 Unauthorized status codes.

CHAPTER 6

DIGITAL SAFETY, CYBERSECURITY PROTOCOLS & KEY GOVERNANCE

Cybersecurity Protocols in Modern Web Engineering

While traditional industrial training often emphasizes mechanical factory safety, software engineering requires rigorous adherence to digital safety, user privacy, and cybersecurity protocols. At Elite Forums, interns were trained to treat application security as a core architectural requirement rather than an afterthought.

Key Security Safeguards Implemented in ProForge

1. API Key Isolation & Environment Variables: Sensitive credentials—including Supabase database service keys, Groq API tokens, and Zoho SMTP passwords—were strictly isolated within .env files and excluded from GitHub using .gitignore rules.
2. Input Sanitization & Injection Prevention: All user inputs across resume textboxes and email forms are sanitized to neutralize Cross-Site Scripting (XSS) and SQL injection payloads.
3. JWT Authorization Guards: Sensitive backend routes are protected with requireAuth middleware, which validates cryptographic signatures before granting access to user data.
4. Multi-Device Login Detection (loginHistory.js): An automated security auditor logs the IP address and User-Agent signature of every login, notifying users if an unrecognized device accesses their account.

CHAPTER 6 (CONTINUED)

BACKUP SYSTEMS, ERROR RECOVERY & PHYSICAL ERGONOMICS

System Backup & Disaster Recovery Protocols

To prevent data loss and ensure uninterrupted development workflows, systematic backup and recovery protocols were strictly followed throughout the internship:

- **Git Remote Redundancy:** All code updates were committed with semantic messages and pushed daily to remote GitHub repositories, allowing instant rollback to any stable previous state.
- **Database Snapshotting:** Supabase PostgreSQL tables and authentication schemas were protected by automated cloud snapshots.
- **Graceful API Degradation:** When external AI services encounter rate limits or transient network failures, backend try-catch handlers fall back to secondary models or offline heuristic parsers.

Physical Ergonomics & Workplace Safety

During on-site training sessions at the Elite Forums center in Vasai, interns followed standard workplace safety and ergonomic guidelines:

- **Ergonomic Posture:** Maintaining proper chair height and monitor distance to reduce spinal strain during extended programming sessions.
- **20-20-20 Eye Rest Rule:** Taking brief breaks every 20 minutes to reduce digital eye strain.
- **Electrical & Fire Safety:** Ensuring all computer power cables, extension strips, and workstation equipment adhere to standard electrical safety and emergency fire protocols.

CHAPTER 6 (CONTINUED)

ETHICAL AI PRINCIPLES, FACTUAL INTEGRITY & ANTI-SPAM STANDARDS

Ethical Governance in AI-Powered Career Platforms

The application of Artificial Intelligence to recruitment and hiring introduces significant ethical responsibilities. During the internship, we held dedicated discussions on AI ethics, ensuring that ProForge was designed with fairness, transparency, and integrity at its core:

1. **Factual Integrity & Anti-Hallucination:** AI prompts inside Remo AI and Covo AI are constrained to refine, polish, and quantify the candidate's genuine experiences. The AI is explicitly instructed never to fabricate false work histories, degrees, or certifications.
2. **User Privacy & Data Ownership:** Candidate resume profiles are strictly private to the user. No personal profile data is sold, monetized, or used to train third-party public AI models without explicit consent.
3. **Ethical Outreach & Anti-Spam (Mali AI):** Recruiter outreach emails generated by Covo AI and scheduled by Mali AI adhere to anti-spam best practices, including clear sender identities, professional subject lines, and transparent intent.
4. **Transparent ATS Matching (Talo AI):** Rather than using deceptive 'white font keyword stuffing' techniques, Talo AI provides honest, constructive guidance on genuine skill gaps.

CHAPTER 7

PRACTICAL EXPERIENCES IN SOFTWARE PRODUCTION (THE 6 AI SUB-SUITES)

Full-Cycle Software Engineering Exposure

During my 12-week internship, I gained hands-on practical experience across the complete software production cycle, architecting and building the six core sub-suites of ProForge:

- 1. Remo AI (Resume Studio): Built an intelligent resume editor with sub-second AI bullet point polishing, real-time live preview, 105+ design combinations, and server-side vector PDFKit rendering.
- 2. Folio AI (Portfolio Generator): Engineered a dynamic portfolio creator supporting four customizable themes (Bento Grid, Cyber Terminal, Modern Executive, Clean Glassmorphism) with client-side ZIP packaging via JSZip.
- 3. Talo AI (ATS Alignment Auditor): Developed a semantic comparison engine that parses job descriptions against candidate resumes to compute percentage match scores and detect missing competencies.
- 4. Covo AI (Outreach Studio): Built a cold outreach generator creating tailored recruiter emails, LinkedIn connection notes, and customized PDF cover letters.
- 5. Liko AI (Personal Branding Architect): Created an AI tool converting project milestones into engaging LinkedIn posts with customizable tone hooks and hashtag suggestions.
- 6. Mali AI (Campaign & Scheduling Studio): Engineered a visual HTML email composer with 12 Google fonts, 7 theme palettes, and an automated background queue scheduler (emailScheduler.js).

CHAPTER 7 (CONTINUED)

SYSTEM ASSEMBLY, API ROUTING & MIDDLEWARE INTEGRATION

Full-Stack System Assembly & API Integration

The system assembly phase involved connecting decoupled frontend React views with backend Express services and cloud databases into a cohesive, high-performance architecture:

- **Modular Express Routing:** Organized backend logic into dedicated route controllers: `analyzeRoutes.js` (AI text parsing), `profileRoutes.js` (database CRUD), `generateRoutes.js` (PDFKit document generation), `authRoutes.js` (OTP verification), and `maliRoutes.js` (email campaign management).
- **Supabase Cloud Database Connection:** Connected backend services to Supabase PostgreSQL using connection pooling and JSONB data types to store dynamic profile objects.
- **Transactional Email Integration:** Configured Nodemailer with Zoho SMTP credentials to handle transactional 6-digit OTP delivery with high deliverability.
- **Zero-Dependency Client Bundling:** Integrated JSZip and FileSaver.js to package standalone portfolio websites directly within the client browser without consuming server bandwidth.

CHAPTER 7 (CONTINUED)

MAINTENANCE, MEMORY PROFILING & PERFORMANCE TUNING

Routine Maintenance & System Optimization

Software maintenance is an essential ongoing activity in production web environments. During the final phase of development, I performed key maintenance and optimization tasks:

- **NPM Package Auditing:** Audited all frontend and backend dependencies using npm audit, upgrading outdated libraries and resolving security vulnerabilities.
- **Background Queue Maintenance:** Optimized emailScheduler.js to run periodic database polling intervals without creating memory leaks or exhausting database connection limits.
- **PDF Memory Buffer Management:** Configured PDFKit to stream binary chunks directly into Express HTTP response streams, significantly reducing RAM usage during concurrent PDF exports.
- **Frontend Bundle Tree-Shaking:** Configured Vite production build settings to eliminate unused code, reducing initial page load times to under 1.2 seconds.

CHAPTER 8

12-WEEK TASK BREAKDOWN (WEEKS 1 TO 4: FOUNDATIONS & CORE)

Week 1: Web Foundations, Modern JavaScript ES6 & Git Version Control

The internship began with an intensive review of modern web standards, HTML5 semantic elements, CSS box sizing, and advanced JavaScript ES6+ features (arrow functions, destructuring, Promises, async/await). We configured professional development environments in Visual Studio Code and mastered Git version control workflows (branching, merging, commit hygiene).

Week 2: React 18 Component Architecture & Vite Tooling

Week two focused on building dynamic Single Page Applications (SPAs) with React 18 and Vite. We mastered component hierarchy, state management using useState and useEffect, custom React hooks, and responsive utility styling using Tailwind CSS v3.

Week 3: Backend Architecture with Node.js & Express REST APIs

In week three, we engineered backend services using Node.js and Express. Key topics included request-response lifecycles, middleware pipelines, CORS configuration, JSON body parsing, and RESTful routing conventions.

Week 4: Cloud Database Setup with Supabase & OTP Authentication

Week four covered cloud database modeling with Supabase PostgreSQL. We designed JSONB schema structures for storing candidate profile data and implemented passwordless email authentication using Zoho SMTP 6-digit OTP verification.

CHAPTER 8 (CONTINUED)

12-WEEK TASK BREAKDOWN (WEEKS 5 TO 8: REMO AI & FOLIO AI)

Weeks 5–6: Remo AI Resume Studio & PDFKit Vector Engine

Weeks five and six marked the development of Remo AI. We engineered the interactive resume editor and connected it to Groq SDK (Qwen 2.5 & Llama 3 models) for real-time bullet point refinement. On the backend, we built a coordinate-based rendering engine in PDFKit that maps content onto strict A4 page boundaries with dynamic height calculation.

Week 7: Folio AI Portfolio Generator & JSZip Packaging

Week seven focused on developing Folio AI. We designed four customizable portfolio themes (Bento Grid, Cyber Terminal, Modern Executive, Clean Glassmorphism) and integrated JSZip for zero-dependency client ZIP downloads containing standalone HTML, CSS, and JS.

Week 8: Talo AI ATS Alignment Engine & Keyword Auditing

In week eight, we built Talo AI, an ATS comparison engine that evaluates candidate resumes against target job descriptions to compute percentage match scores and highlight missing technical keywords.

CHAPTER 8 (CONTINUED)

12-WEEK TASK BREAKDOWN (WEEKS 9 TO 11: COVO, LIKO & MALI AI)

Week 9: Covo AI Outreach Studio & PDF Cover Letter Generator

During week nine, we developed Covo AI, creating personalized recruiter cold emails, LinkedIn connection messages, and downloadable PDF cover letters formatted with professional headers and sign-offs.

Week 10: Liko AI LinkedIn Personal Branding Architect

Week ten focused on Liko AI, creating a personal branding tool that transforms career achievements into engaging LinkedIn posts with customizable hooks, hashtags, and optimized profile bios.

Week 11: Mali AI Campaign Studio & Background Queue Scheduler

In week eleven, we developed Mali AI, building a visual HTML email composer with 12 Google fonts, 7 theme palettes, dual time pickers, and an automated background queue worker (emailScheduler.js).

CHAPTER 8 (CONTINUED)

12-WEEK TASK BREAKDOWN (WEEK 12: SECURITY & FINAL DEMO)

Week 12: Cybersecurity Hardening, Multi-Theme System & Final Evaluation

The final week focused on cybersecurity auditing, multi-theme visual polish, and technical project defense. We integrated an active security monitoring system (loginHistory.js) logging IP addresses and User-Agent signatures to issue multi-device alerts.

We also finalized the 10-theme visual design system with smooth cubic-bezier transitions, performed end-to-end regression testing, and successfully defended the ProForge AI platform during the final evaluation at Elite Forums.

Summary of Core Technical Deliverables Built During Training

- Remo AI: Resume Studio with 105+ layout options and strict A4 PDF compilation.
- Folio AI: 4 responsive portfolio templates with zero-dependency ZIP export.
- Talo AI: ATS alignment engine with keyword gap analysis.
- Covo AI: Cold outreach studio & dynamic cover letter generator.
- Liko AI: LinkedIn post generator & bio optimizer.
- Mali AI: Email campaign composer with background scheduling queue.

CHAPTER 9

TECHNICAL CHALLENGES & ENGINEERING SOLUTIONS (PDF & LLM)

1. Dynamic Coordinate Math & A4 Pagination in PDFKit

Challenge Encountered: Unlike web browsers where CSS reflows content automatically, PDFKit operates on strict (X, Y) pixel coordinates on a fixed A4 canvas. When users entered lengthy experience descriptions, text initially spilled over the bottom page boundary or broke awkwardly across pages.

Engineering Solution: I built a pre-render height calculation utility using `doc.heightOfString()` to measure every block of text before rendering. If an entry would push past the safe boundary (780 pt), the engine triggers `doc.addPage()` and recalculates vertical offsets cleanly.

2. Schema Enforcement from Non-Deterministic LLM Outputs

Challenge Encountered: During early Groq SDK testing, AI models occasionally returned conversational explanations or markdown code blocks instead of clean JSON, causing backend JSON parsing errors.

Engineering Solution: I enforced `response_format: { type: 'json_object' }` in Groq API calls and added a backend sanitization utility that strips markdown wrappers, guaranteeing 100% valid JSON objects.

CHAPTER 9 (CONTINUED)

BACKGROUND QUEUEING, SPRINT COORDINATION & STUDENT LEARNINGS

3. Asynchronous Background Scheduling for Email Campaigns

Challenge Encountered: Building Mali AI's campaign scheduler required running automated background dispatches without blocking the Node.js event loop or requiring heavy third-party Redis queue servers.

Engineering Solution: I created a lightweight polling queue worker inside `backend/services/emailScheduler.js` that runs on periodic intervals, checking for pending campaigns and dispatching them via Zoho SMTP safely.

Key Student Learnings & Professional Reflections

As a diploma student completing this training under the MSBTE curriculum, this internship provided immense technical and personal growth:

- **Full-Stack Mastery:** Transitioned from writing basic academic code to architecting production-grade React, Node.js, and PostgreSQL applications.
- **Debugging & Analytical Thinking:** Developed structured debugging skills using browser developer tools, Postman, and server logs.
- **Time Management:** Balanced daily on-site training sessions with independent capstone development across 12 intensive weeks.

CHAPTER 10

CONCLUSION, INDUSTRY EXPERIENCE & FUTURE SCOPE

My 12-week industrial training at Elite Forums, Vasai (East), has been an invaluable learning experience that successfully fulfilled all MSBTE Diploma curriculum objectives for Computer Engineering. The program provided a perfect bridge between theoretical academic concepts and live software industry practices.

Developing the ProForge AI platform enabled me to gain deep, hands-on expertise in React 18, Node.js, Express, Supabase PostgreSQL, Groq SDK LPU inference, and server-side PDFKit vector compilation. Overcoming complex engineering challenges—such as multi-model AI fallbacks and strict A4 coordinate math—significantly enhanced my problem-solving abilities and confidence.

Future Enhancements for ProForge AI

- Multi-Language Translation: Supporting automated resume translation into major Indian and international languages.
- AI Voice Mock Interviews: Building interactive voice-based mock interview simulations tailored to specific job descriptions.
- Placement Cell Portals: Developing multi-tenant institutional dashboards for polytechnic and engineering colleges.

In conclusion, this industrial training has equipped me with practical software engineering skills that will serve as a strong foundation for my future career in computer engineering.

CHAPTER 11

REFERENCES, TECHNICAL DOCUMENTATION & CORPORATE INFORMATION

1. Official Documentation & Technical Standards

- React 18 Official Documentation – <https://react.dev/>
- Vite Frontend Tooling Guide – <https://vitejs.dev/>
- Tailwind CSS Design System Documentation – <https://tailwindcss.com/docs>
- Node.js Event Loop & Architecture Reference – <https://nodejs.org/docs>
- Express.js Routing & Middleware Reference – <https://expressjs.com/>
- Groq SDK LPU Inference & API Guide – <https://console.groq.com/docs>
- PDFKit Vector Document Generation Reference – <https://pdfkit.org/docs/>
- Supabase Database & Auth Documentation – <https://supabase.com/docs>
- Nodemailer Transport Client Documentation – <https://nodemailer.com/>
- JSZip Client-Side Zip Library – <https://stuk.github.io/jszip/>

2. Educational References & Textbooks

- JavaScript: The Definitive Guide – David Flanagan, O'Reilly Media
- Learning React: Modern Patterns for Developing React Apps – Alex Banks & Eve Porcello
- MSBTE Industrial Training Curriculum Guidelines (Computer Engineering Group)

Training Industry: Elite Forums

<https://in.linkedin.com/company/eliteforums>

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